SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Coding Challenge Task

Course Code:	CSA0615	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	**CO4:** Articulation (BL4, BL5)	**Assessment:**	Assessment Tool 1 – Coding Challenge Task
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	V.LASYA	**Reg. No.:**	192524186
Section / Batch:	B.TECH AI&DS	**Date:**	21-08-2026

Instructions to Students

- Answer both questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given questions.
- Write clear code for the given problem.
- Analyze the Time complexity and Space complexity of your code using dynamic programming and greedy strategies

Question Type	Description	Questions	Total Marks
Coding Challenge Task	Focus on Code and analyze optimization problems using dynamic programming and greedy strategies	Q1 – Q2	100

RUBRICS FOR CALCULATION

Coding Challenge Task

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

SECTION A – Coding Challenge Task

SIMATS ENGINEERING

SIMATS Online Programming Lab

DAA PROGRAMMING

Vandana Lasya
192524186RA

CO4-AT1-Coding Challenge

Back

Language: Python | C | C++ | Java

QUESTIONS

Q1
Dynamic Programming
50 marks

Q2
Dynamic Programming
50 marks

1/2 answered

Q1 Dynamic Programming 50 marks

Partition Equal Subset Sum

Given an array of positive integers, the task is to determine whether the array can be divided into two subsets such that both subsets have the same sum. Each element must belong to exactly one subset. The goal is to check whether such an equal partition is possible. Use Dynamic Programming to determine the feasibility of the partition.

Input Format:
n = number of elements
arr[] = array elements
Sample Input:
n = 4
arr = 1 5 11 5
Expected Output:
Partition Possible = Yes

Your Code

```python
n=int(input())
arr=list(map(int,input().split()))
total=sum(arr)
if total%2!=0:
    print("partition possiable=No")
else:
  target=total//2
  dp=[False]*(target+1)
  dp[0]=True
  for num in arr:
    for j in range(target,num-1,-1):
      dp[j]=dp[j]or dp[j-num]
if dp[target]:
  print("partiton possible=yes")
else:
  print("partiton possiable = No")
```

Input

```
4
1 5 11 5
```

Output

```
partiton possible=yes
```

Submitted

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CO4-AT1-Coding Challenge

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Language: Python | C | C++ | Java

QUESTIONS

Q1
Dynamic Programming
50 marks

Q2
Dynamic Programming
50 marks

2/2 answered

Q2 Dynamic Programming 50 marks

Maximum Length Chain of Pairs

Given pairs of numbers where each pair has a start and end value, the goal is to form the longest chain such that the end of one pair is less than the start of the next. Efficient selection is required to maximize chain length. Use Greedy or Dynamic Programming techniques.

Input Format:
n = number of pairs
pairs[][]
Sample Input:
n = 3
(5,24) (15,25) (27,40)
Expected Output:
Max Chain Length = 2

Your Code

```
n=int(input())
pairs=[]
for _ in range(n):
  a,b = map(int,input().split())
  pairs.append((a,b))
pairs.sort()
dp=[1]*n
for i in range(n):
  for j in range(i):
    if pairs[j][1]<pairs[i][0]:
       dp[i]=max(dp[i],dp[j]+1)
print("Max Chain Length =",max(dp))
```

Input

```
3
5 24
15 25
27 40
```

Output

```
Max Chain Length = 2
```

Submitted

Code of Conduct

I ____V,LASYA(192524186)______________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V.lasya ______________________

Mark Evaluation:

Questions	Problem Understanding (20)	Algorithm (25)	Code Implementation (20)	Competitive Performance (20)	Test Case Handling (15)	Total Marks (Each - 50)
Q1						
Q2						
					Overall Total	

Faculty In-charge	Course Coordinator	Head of Department
Name: ________________ Signature: ________________	Signature: ________________	Signature: ________________
Date: ________________	Date: ________________	Date: ________________